Blackbody Radiation

Laboratory Report

Parth
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1 Aim

To study the radiation emitted by the incandescent tungsten filament of a bulb, treated as a blackbody, by

1. determining the filament temperature at several bulb voltages from its electrical resistance,
2. measuring the intensity of the emitted light as a function of angular position with a prism spectrophotometer, converting angle to wavelength and finding the wavelength $\lambda_{\max}$ of maximum intensity,
3. verifying Stefan's law ($I \propto T^4$), and
4. verifying Wien's displacement law ($\lambda_{\max} T = \text{const.}$) and estimating Wien's constant.

2 Theory

2.1 Blackbody radiation

A blackbody is an ideal body that absorbs all radiation falling on it. In thermal equilibrium it must also emit as efficiently as it absorbs, and the spectrum it emits depends *only* on its temperature and not on its material or shape. A small hole in a closed cavity whose walls are held at temperature T is the standard realisation. Classical physics failed to explain the measured spectrum: the Rayleigh–Jeans law, derived from equipartition of energy among the electromagnetic modes of the cavity, gives a spectral radiance

$$B_\lambda^{\mathrm{RJ}}(T) = \frac{2ck_BT}{\lambda^4}, \tag{1}$$

which agrees with experiment at long wavelengths but diverges as $\lambda \to 0$ (the "ultraviolet catastrophe") and predicts an infinite total power. In 1900 Max Planck removed the divergence by postulating that the cavity oscillators can exchange energy with the radiation field only in discrete units $E = h\nu$, which marks the beginning of quantum mechanics.

2.2 Planck's law

With the energy of an oscillator of frequency ν restricted to $E_n = nh\nu$, its mean energy at temperature T is $\bar{E} = h\nu/(e^{h\nu/k_BT} - 1)$ instead of the classical k_BT. The spectral radiance (power per unit area, per unit solid angle, per unit wavelength) is then

$$B_\lambda(T) = \frac{2hc^2}{\lambda^5}\,\frac{1}{e^{hc/\lambda k_BT} - 1}, \tag{2}$$

where $h = 6.626 \times 10^{-34}\ \mathrm{J\,s}$, $c = 2.998 \times 10^{8}\ \mathrm{m\,s^{-1}}$ and $k_B = 1.381 \times 10^{-23}\ \mathrm{J\,K^{-1}}$. For $\lambda \gg hc/k_BT$ Eq. (2) reduces to the Rayleigh–Jeans law, and for $\lambda \ll hc/k_BT$ to Wien's approximation

$B_\lambda \approx (2hc^2/\lambda^5)\, e^{-hc/\lambda k_B T}$. Figure 1 shows the black-body spectrum for several temperatures: the curve rises steeply on the short-wavelength side, passes through a single maximum and falls slowly on the long-wavelength side; as T increases the total area grows rapidly and the peak moves to shorter wavelengths.

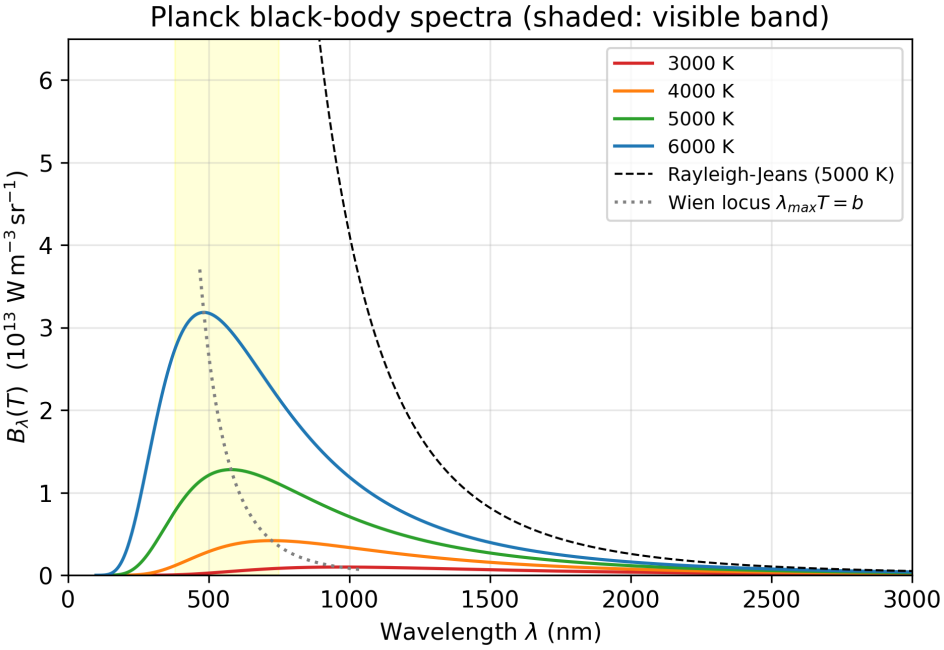


Figure 1: Black-body spectra from Planck's law for four temperatures, the classical Rayleigh–Jeans curve (5000 K) and the locus of the maxima (Wien's law).

2.3 Stefan's law

Integrating Eq. (2) over all wavelengths and over the hemisphere gives the total power radiated per unit area of a blackbody,

$$M = \int_0^\infty \pi B_\lambda \, \mathrm{d}\lambda = \sigma T^4, \qquad \sigma = \frac{2\pi^5 k_B^4}{15\, c^2 h^3} = 5.67 \times 10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}}. \tag{3}$$

Hence the total detected intensity should obey $I_{\mathrm{tot}} = kT^4$, i.e.

$$\ln I_{\mathrm{tot}} = 4 \ln T + \ln k, \tag{4}$$

so a plot of $\ln I_{\mathrm{tot}}$ against $\ln T$ is a straight line of slope ≈ 4. A real body with emissivity $\varepsilon < 1$ (tungsten has $\varepsilon \approx 0.3$–0.4 in this range) radiates $\varepsilon\sigma T^4$; if ε varies only slowly with T the slope is still close to 4.

2.4 Wien's displacement law

Setting $\mathrm{d}B_\lambda/\mathrm{d}\lambda = 0$ and writing $x = hc/\lambda k_B T$ gives the transcendental equation $(x-5)e^x + 5 = 0$, whose root is $x = 4.965$. The wavelength of maximum emission therefore satisfies

$$\lambda_{\max} T = b, \qquad b = \frac{hc}{4.965\, k_B} = 2.898 \times 10^{-3}\ \mathrm{m\,K}. \tag{5}$$

A graph of $\lambda_{\max}$ against $1/T$ is a straight line through the origin with slope b.

2.5 The filament as a blackbody and its temperature

The filament of an incandescent bulb is a hot, opaque, tightly coiled tungsten wire; the coil traps most of its own radiation (cavity-like) so its spectrum is close to that of a (grey) blackbody of the filament temperature. The temperature follows from the temperature dependence of the resistance of tungsten,

$$R = R_0\,[1 + \alpha_0\,(T - T_0)] \quad \Longrightarrow \quad T = T_0 + \frac{1}{\alpha_0}\left(\frac{R}{R_0} - 1\right), \tag{6}$$

where R_0 is the resistance at room temperature T_0 and $\alpha_0 = 4.5 \times 10^{-3}\ \mathrm{K}^{-1}$. (The handout prints a minus sign in the bracket; this is a typographical error, since the resistance of a metal increases with temperature.) R_0 is found from the slope of the V–I graph at a voltage so small that the filament does not heat up, and $R = V/I$ at the working voltage.

2.6 Wavelength from the prism deviation angle

The 60° prism disperses the light, and the Rotary Motion Sensor records the angle θ of the emergent ray measured from the undeviated beam (θ_0). With the beam incident at 60° on the first face of a prism of apex angle 60°, the refractive index is

$$n = \sqrt{\left(\frac{2}{\sqrt{3}}\sin\theta + \frac{1}{2}\right)^2 + \frac{3}{4}}, \tag{7}$$

and the dispersion of the glass is described by Cauchy's relation

$$n = \frac{A}{\lambda^2} + B, \qquad A = 13\,900\ \mathrm{nm}^2, \quad B = 1.689, \tag{8}$$

so that

$$\lambda = \sqrt{\frac{A}{n - B}}\ \ (\mathrm{nm}). \tag{9}$$

Equation (9) is defined only for $n > B$, i.e. for $\theta > 55.37°$; wavelengths longer than about 1.3 μm are squeezed into less than 1° of arm angle.

3 Calculations and Results

3.1 Resistance of the filament at room temperature

With small applied voltages (0.01–0.10 V, maximum dissipation 2.7 mW) the filament stays at room temperature and obeys Ohm's law. Figure 2 shows the V–I graph. A least-squares fit $V = R_0 I + c$ gives

$$R_0 = 3.62 \pm 0.09\ \Omega, \qquad c = 0.001 \pm 0.002\ \mathrm{V}\ \text{(consistent with zero)}, \qquad r = 0.998. \tag{10}$$

A fit forced through the origin gives $R_0 = 3.68 \pm 0.04\ \Omega$, and the value noted in the laboratory notebook is 3.66 Ω; all three agree within the uncertainty. The fitted slope $R_0 = 3.62\ \Omega$ is used below. The room temperature was not recorded and is taken as $T_0 = 300 \pm 3$ K.

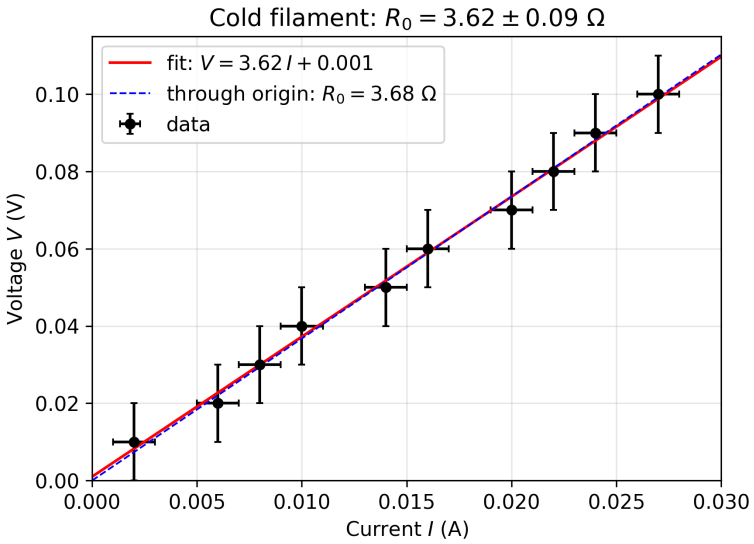


Figure 2: Voltage against current for the cold filament; the slope gives R_0.

3.2 Temperature of the filament

At each bulb voltage the current was noted and $R = V/I$ and T were obtained from Eq. (6). *Worked example, 6.00 V:*

$$R = \frac{6.00}{0.256} = 23.44\ \Omega, \qquad \frac{R}{R_0} = \frac{23.44}{3.62} = 6.47,$$
$$T = 300 + \frac{6.47 - 1}{4.5 \times 10^{-3}} = 300 + 1215 \approx 1515\ \text{K}.$$

Run	V (V)	I (A)	$P = VI$ (W)	R (Ω)	T (K)	T (°C)
5	6.0	0.256	1.54	23.44 ± 0.10	1515 ± 36	1242
6	7.0	0.280	1.96	25.00 ± 0.10	1611 ± 39	1337
12	7.5	0.290	2.18	25.86 ± 0.10	1663 ± 40	1390
9	8.0	0.300	2.40	26.67 ± 0.09	1713 ± 41	1440

Table 1: Filament resistance and temperature at the four bulb voltages used. (Readings at 6.5 V and 9 V were struck out in the notebook and are not used.)

3.3 Intensity as a function of angular position

The angular position was zeroed with the undeviated white beam centred on the aperture slit ($\theta_0 = 0$), so the emergent angle is $\theta = |\text{reading}|$. The sensor values are taken directly as the arm angle in degrees (the calibration run, which records the angle channel against the calibrated angle channel, gives a straight line of slope 1.000 and intercept 0, so no rescaling was needed). The recorded intensity is a percentage of the full scale of the light sensor, and each run has its own constant offset (-7% to $+1\%$) because the sensor zero was re-set between runs.

Figure 3 shows the intensity against angle for the four voltages. Away from the feature the signal is flat or drifts slowly (stray and room light). Between $\theta \approx 56°$ and $63°$ there is a single smooth peak that grows with the bulb voltage and moves slightly towards larger angles. For the analysis a straight-line baseline was fitted to line-free windows on either side (53°–55.8° and 64.5°–68°; rms residual 0.05–0.07 %) and subtracted (Fig. 3b).

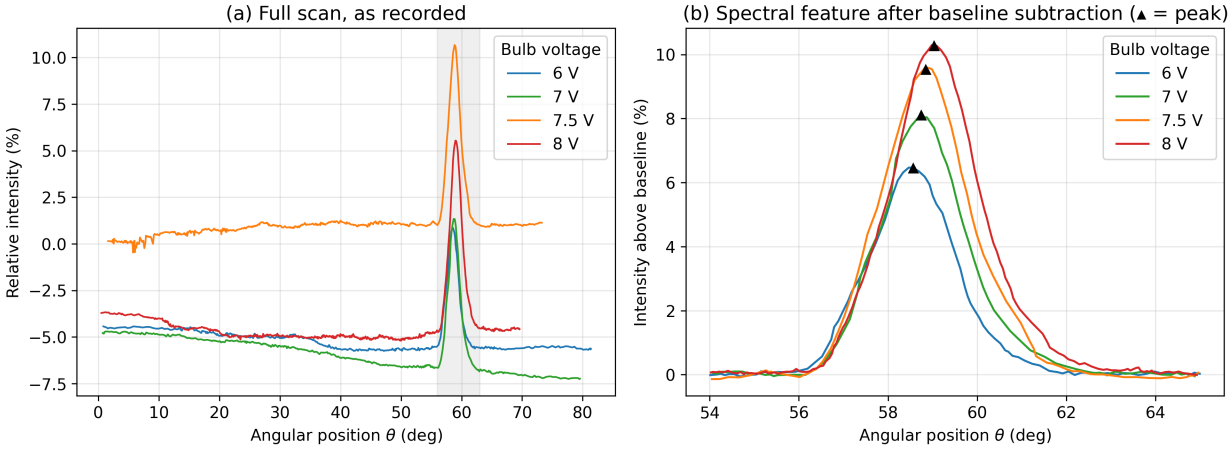


Figure 3: Relative intensity against angular position for bulb voltages of 6, 7, 7.5 and 8 V. (a) Full scans as recorded (shaded band: region of the spectral feature). (b) The feature after baseline subtraction; triangles mark the fitted peak positions.

3.4 Intensity as a function of wavelength and $\lambda_{\max}$

Each angle was converted to a wavelength with Eqs. (7)–(9). *Worked example, 6.0 V:* the peak is at $\theta = 58.57^\circ$ (vertex of a parabola fitted to the points within $\pm 0.7^\circ$ of the maximum), so

$$n = \sqrt{\left(\tfrac{2}{\sqrt{3}}\sin 58.57^\circ + \tfrac{1}{2}\right)^2 + \tfrac{3}{4}} = \sqrt{(0.9853 + 0.5)^2 + 0.75} = 1.7193,$$

$$\lambda_{\max} = \sqrt{\frac{13\,900}{1.7193 - 1.689}} = \sqrt{\frac{13\,900}{0.0303}} \approx 677 \text{ nm}.$$

The uncertainty $\pm 0.10^\circ$ in θ (fit error combined with the 0.1° reading uncertainty) corresponds to ± 10 nm. Figure 4 shows the baseline-subtracted intensity against wavelength, and Table 2 lists the results.

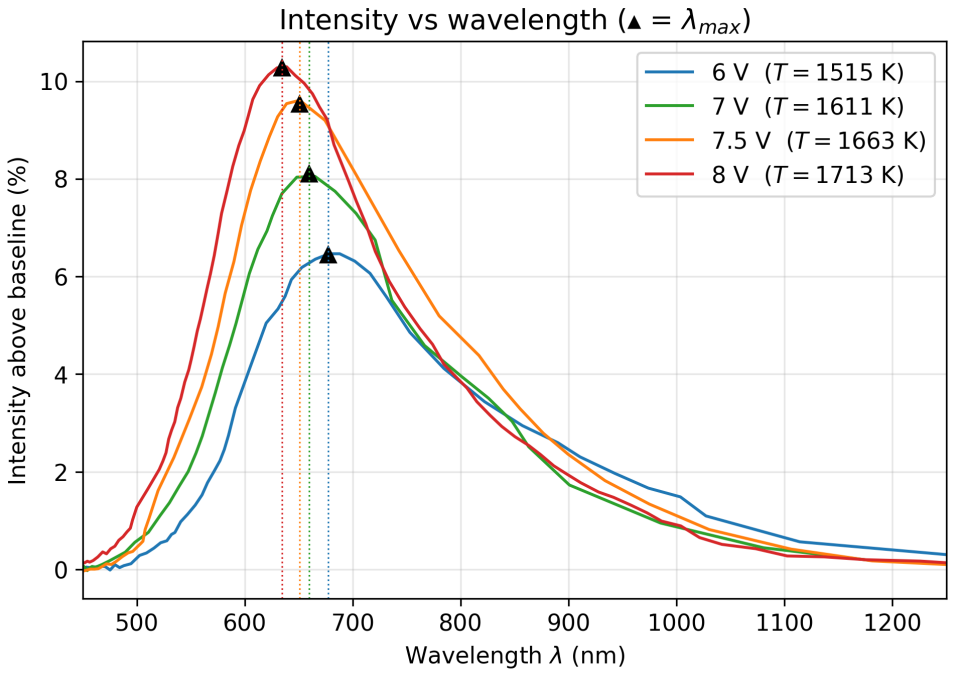


Figure 4: Intensity against wavelength for the four bulb temperatures. The triangles and dotted lines mark $\lambda_{\max}$.

V (V)	T (K)	$\theta_{\max}$ (°)	n	$\lambda_{\max}$ (nm)	$I_{\max}$ (%)	I_{tot} (%·°)	$\lambda_{\max}^{\mathrm{Wien}}$ (nm)
6.0	1515	58.57 ± 0.10	1.7193	677 ± 10	6.45	15.4 ± 0.4	1913
7.0	1611	58.75 ± 0.10	1.7209	660 ± 9	8.10	19.3 ± 0.4	1799
7.5	1663	58.85 ± 0.10	1.7218	651 ± 9	9.53	23.7 ± 0.6	1742
8.0	1713	59.04 ± 0.10	1.7235	634 ± 8	10.27	26.1 ± 0.5	1692

Table 2: Peak position, refractive index, $\lambda_{\max}$, peak height above baseline, integrated intensity $I_{\mathrm{tot}} = \int I \, \mathrm{d}\theta$ over 56°–63°, and, for comparison, the value b/T expected from Wien's law with $b = 2.898 \times 10^{-3}$ m K.

As the filament gets hotter, the peak grows and shifts to *shorter* wavelength (from 677 to 634 nm for a rise of about 200 K), as the blackbody picture predicts. The full width at half maximum covers roughly 560–860 nm in every run. However, the measured peaks (630–680 nm) lie far below the wavelengths $b/T \approx 1.7$–1.9 μm at which a blackbody of this temperature emits most strongly (last column of Table 2); this is discussed in Sec. 5.

3.5 Verification of Stefan's law

The total (baseline-subtracted) intensity I_{tot} was obtained by numerically integrating each spectrum over the whole feature, $56° \le \theta \le 63°$ (the detector collects the energy per unit angle through a slit of fixed angular width, so the integral over θ is proportional to the total detected power). Figure 5(a) is the graph of $\ln I_{\mathrm{tot}}$ against $\ln T$ suggested by Eq. (4), and (b) shows I_{tot} against T^4. The least-squares line gives

$$\text{slope} = 4.43 \pm 0.34 \quad (r = 0.994), \tag{11}$$

which agrees with the expected value of 4 within about 1.3 standard errors. Using the peak height $I_{\max}$ instead of the integral gives 3.90 ± 0.24 ($r = 0.996$), also consistent with 4; the mean of the two estimates is ≈ 4.2. As a cross-check, the electrical power $P = VI$ supplied to the bulb varies as $T^{3.64 \pm 0.13}$; it is somewhat below 4 because part of the input power leaves by conduction through the lead wires and gas and because the emissivity of tungsten increases with temperature.

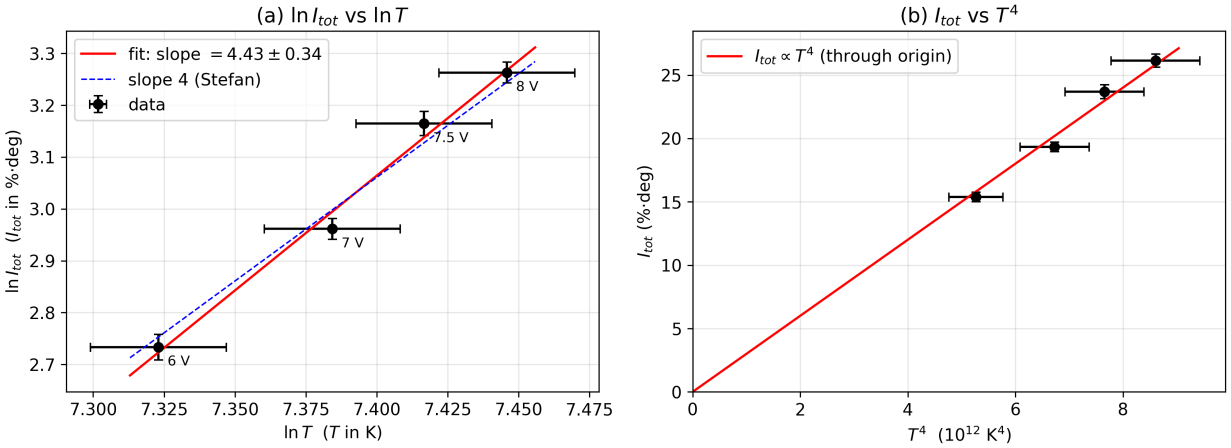


Figure 5: Verification of Stefan's law. (a) $\ln I_{\mathrm{tot}}$ against $\ln T$ with the fitted line and, for comparison, a line of slope exactly 4. (b) I_{tot} against T^4 with a straight line through the origin.

3.6 Verification of Wien's displacement law

Figure 6(a) shows $\lambda_{\max}$ against $1/T$ and (b) the product $\lambda_{\max} T$ against T. Fitting $\lambda_{\max} = b/T$ (line through the origin) gives

$$b = (1.06 \pm 0.01) \times 10^6 \ \text{nm K} = (1.06 \pm 0.01) \times 10^{-3} \ \text{m K}, \tag{12}$$

where the quoted error is the statistical error of the fit only. The product $\lambda_{\max}T$ is nearly constant over the range studied: 1.03, 1.06, 1.08, 1.09 ($\times 10^{-3}$ m K) for 6, 7, 7.5 and 8 V, mean $(1.06 \pm 0.03) \times 10^{-3}$ m K (scatter 2.6 %), i.e. $\lambda_{\max}$ varies approximately as $1/T$ in the sense of Wien's law. A free straight-line fit $\lambda_{\max} = a + s/T$ gives $s = (5.4 \pm 0.7) \times 10^5$ nm K and $a = 325 \pm 44$ nm ($r = 0.983$), and a log–log fit gives $\lambda_{\max} \propto T^{-0.51 \pm 0.06}$ (Wien's law: -1). These two fits are poorly constrained because T spans only 13 % (1515–1713 K) and the uncertainty in T (± 2.4 %) is comparable to the whole change in $\lambda_{\max}$ (-6 %), so they should be regarded as indicative only.

The measured coefficient is $b_{\text{exp}} = 1.06 \times 10^{-3}$ m K, which is only 37 % of the accepted value 2.898×10^{-3} m K (a deviation of -63 %). Wien's law is therefore confirmed in its qualitative form (the peak moves to shorter wavelength as T rises and $\lambda_{\max}T$ is approximately constant), but not quantitatively.

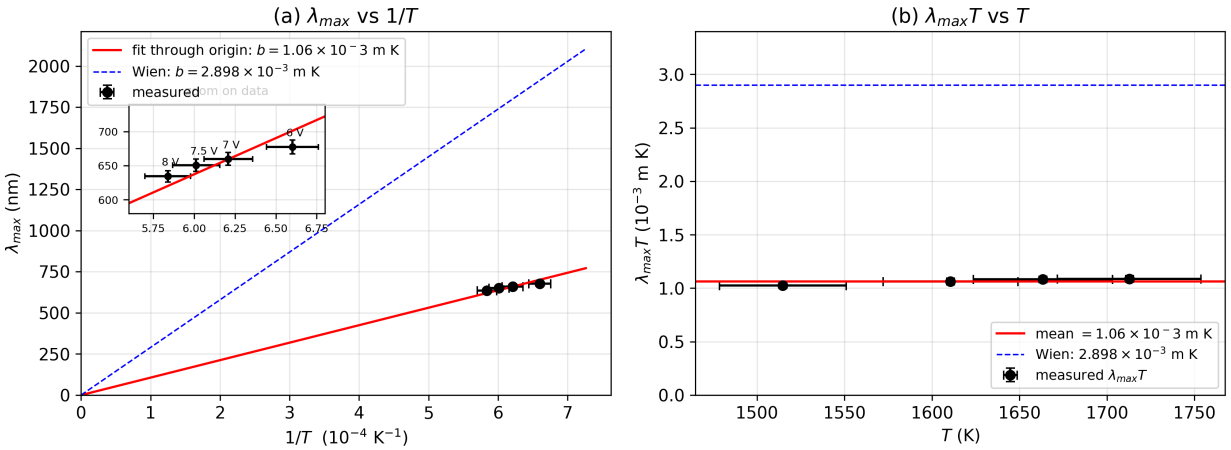


Figure 6: Verification of Wien's law. (a) $\lambda_{\max}$ against $1/T$ with the fit through the origin and the line for the accepted $b = 2.898 \times 10^{-3}$ m K; the inset magnifies the data. (b) $\lambda_{\max}T$ against T.

4 Error Analysis

- **Resistance and temperature.** V and I were read to ± 0.01 V and ± 0.001 A, giving $\Delta R \approx 0.1\ \Omega$ at the working points. The error in R_0 ($\pm 0.09\ \Omega$, 2.5 %, from the fit) and the assumed $\Delta T_0 = 3$ K dominate: combining them with ΔR in quadrature in Eq. (6) gives $\Delta T = 36$–41 K (≈ 2.4 %). There is also an unquantified *systematic* error because Eq. (6) treats α_0 as constant up to ≈ 1700 K and T_0 was not measured. The temperatures are nevertheless consistent with tabulated tungsten resistivity at these temperatures, and since the same R_0, T_0 and α_0 enter all runs they mostly cancel in the slope of the Stefan plot.

- **Angle and wavelength.** The peak position is uncertain by $\pm 0.10°$ (parabola-fit error and 0.1° reading/zero uncertainty), which gives $\Delta\lambda_{\max} \approx 8$–$10$ nm (1.3–1.5 %). Because λ depends strongly on θ (about 90 nm per degree near the peak and increasing rapidly towards 55.4°), any error in the zero θ_0 of the angle scale shifts all $\lambda_{\max}$ together. An error of 1° in θ_0 would change $\lambda_{\max}$ by about 90 nm.

- **Intensity and Stefan's law.** The rms noise of the baseline (0.05–0.07 %) gives $\Delta I_{\text{tot}} \approx 0.4$–$0.6$ %·° (2–3 %). The slope of the Stefan plot is 4.43 ± 0.34 (statistical standard error of the fit, ≈ 8 %); with only four points this is a limited test. The comparison of intensities between runs assumes that the sensor gain was the same in all runs.

- **Wien's constant.** The scatter of $\lambda_{\max}T$ is 2.6 % and the statistical error of b is only 1.3 %, but the systematic deviation of b from the accepted value (63 %) is much larger than any of the random errors estimated above and therefore has a physical origin, discussed in the conclusions.

5 Summary and Conclusions

Result	Value
Cold resistance of filament (from V–I graph)	$R_0 = 3.62 \pm 0.09\ \Omega$
Filament temperature at 6, 7, 7.5, 8 V	1515, 1611, 1663, 1713 K ($\pm$ 36–41 K)
Wavelength of maximum intensity	677, 660, 651, 634 nm ($\pm$ 8–10 nm)
Slope of $\ln I_{\mathrm{tot}}$ vs $\ln T$ (Stefan; expected 4)	4.43 ± 0.34 (from peak heights: 3.90 ± 0.24)
$\lambda_{\max} T$ (mean of four runs)	$(1.06 \pm 0.03) \times 10^{-3}$ m K
Wien's constant b (fit through origin)	$(1.06 \pm 0.01) \times 10^{-3}$ m K (accepted 2.898×10^{-3})

- The V–I graph of the cold filament is linear ($r = 0.998$) and gives $R_0 = 3.62 \pm 0.09\ \Omega$. The resulting filament temperatures, 1515–1713 K for 6–8 V, agree with the range expected for a tungsten bulb running at these voltages.
- The intensity–angle scans show a single feature between 56° and 63° whose height increases and whose position shifts to larger angle (shorter wavelength) with increasing bulb voltage, as expected for blackbody emission.
- **Stefan's law** is verified: the slope of $\ln I_{\mathrm{tot}}$ against $\ln T$ is 4.43 ± 0.34 (and 3.90 ± 0.24 from the peak heights), consistent with the expected value of 4 within the experimental uncertainty.
- **Wien's law** is verified only qualitatively. The peak wavelength decreases with temperature and the product $\lambda_{\max} T$ is constant to within 2.6 % over the range studied, but its value, $b = 1.06 \times 10^{-3}$ m K, is about one third of the accepted 2.898×10^{-3} m K. At 1500–1700 K the true emission maximum lies at 1.7–1.9 μm, in the infrared, where the apparatus is least effective. The most likely causes are (i) the measured curve is the product of the blackbody spectrum and the spectral response of the apparatus (the transmission of the prism and lenses and the responsivity of the light sensor), which drops towards the infrared, so the observed peak lies where this product is largest and shifts only weakly with T; (ii) the sensor measures power per unit angle rather than per unit wavelength and the dispersion becomes extremely non-linear beyond ≈ 1 μm, which distorts the spectrum on the long-wavelength side. No correction was made for either effect. The Wien coefficient obtained here is therefore a property of this instrument–source combination and not an estimate of the universal constant; the experiment would need a source at a higher temperature (peak in the visible) or a calibrated instrument response to reproduce 2.898×10^{-3} m K.
- Only four temperatures were available (the 6.5 V and 9 V measurements were discarded in the lab), and the temperature range is narrow, so the quantitative tests rely on four points only.

Sources of error. The room temperature T_0 was not recorded; α_0 is assumed constant up to ≈ 1700 K; the filament is not a perfect blackbody (tungsten emissivity < 1 and wavelength dependent); the wavelength scale depends on the Cauchy constants A, B and on the assumed zero of the angle scale; the spectral response of the prism and light sensor was not corrected; stray and room light produced a drifting baseline; and the sensor zero and offset were changed between runs.